

Smooth source, unstructured head: a critical synthesis

The shared problem is how to generate plausible future patches from a past context. A source distribution supplies the initial curve; a learned, time-dependent vector field transports it toward the conditional future distribution. Our question concerns their interaction: does replacing white noise by a smooth Gaussian-process (GP) source make an ordinary adaptive-layer-normalized MLP harder to train, and does temporal structure or whitening remedy any gap? This is a hypothesis to test, not an established defect of flow matching.

TimeFlow in Sundial. The published formulation uses a Gaussian white-noise source and a small MLP conditioned on a Transformer representation, with adaptive layer normalization and multi-patch prediction. Its evaluation concerns a pretrained forecasting system: for example, GIFT-Eval uses 23 datasets, 97 configurations, 100 generated paths, MASE and CRPS. These results establish forecasting performance for that complete system, not the suitability of its head for a matched-GP source. White noise to a smooth target is therefore the published reference geometry; it is not the disputed smooth-source cell. [Liu et al., 2025, §§4.1.3, 5.1.2 and Table 2.](#)

TSFlow. This method explicitly studies GP priors and uses a DiffWave-style network with S4 temporal layers. It also considers optimal-transport coupling and conditional prior sampling. Its eight-dataset forecasting evaluation estimates CRPS using nine quantiles and 100 samples, reports five seeds, and uses Euler integration. Prior ablations within this architecture provide evidence that the source can help that network. They do not settle whether an unstructured MLP handles the same source equally well: TSFlow is not a direct counterexample to the head hypothesis until architecture and experimental conditions match. Our independent-coupling, tiny-encoder experiment deliberately removes several features of the full TSFlow system. [Kollovieh et al., 2025, §§3-4.](#)

Why S4 is relevant. S4 parameterizes a structured state-space sequence transformation; its temporal inductive bias gives a concrete alternative to patchwise dense regression. Its long-sequence results motivate this comparison but do not imply an advantage at horizon 16. Testing that advantage is necessary. The official implementation also assigns special optimizer settings to state-space parameters, which should be preserved or deviations disclosed. [Gu, Goel and Ré, 2022; official S4 implementation.](#)

The missing comparison. Cross white-noise and matched-GP sources with AdaLN MLP and three-block S4 heads, holding data, encoder design, training opportunity and evaluation draws fixed. Confirm the target and velocity spectra first. Then measure leading-PC velocity error, residual-PC error and generated-path roughness; run a GP-source whitened MLP as the required control. Unlike separate paper summaries, this design distinguishes a source effect from an architecture effect within one controlled setting.

Limits on interpretation. Low-rank velocity covariance alone does not prove poor optimization. Independent-coupling velocity targets retain regression noise, so pairwise MSE includes an irreducible component; the optimal regressor is the conditional mean velocity. Whitening rescales directions and the loss, rather than necessarily removing dimensions. An S4 advantage would support a local design recommendation, not establish its cause or show that pretrained Sundial fails. Absence of an MLP disadvantage is an equally valid outcome. [Flow Matching, Lipman et al., 2023.](#)